MILITARY SPECIFICATION

BOLT, TEE

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of bolt entitled Bolt, Tee, which is used to attach the Lance warhead to the missile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-I-6868 - Inspection Process, Magnetic Particle

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipment.
8808932 - Packing for Hardware and Related Items.
9237322 - Bolt, Tee.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The bolt shall comply with all requirements specified on the drawing (dwg) 9237322, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Hardness. The Rockwell hardness of the bolt shall be in the range of C35 to C40 when tested in accordance with 4.4.1.

3.4 Plating thickness. The plating thickness shall be 0.0002 to 0.0004 inch thick when tested in accordance with 4.4.2.

3.5 Tensile load. The bolt shall withstand a tensile load when tested in accordance with 4.4.3. There shall be no cracks observed visually or by magnetic particle inspection (MIL-I-6868), and there shall be no deformation of the threads.

3.6 Continuous grain flow lines. The bolt shall conform to the grain flow lines as shown in Figure 2 when tested in accordance with 4.4.4.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection. Not applicable.

4.2.1 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.2 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Bolt, tee (see dwg 9237322).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Evidence of baking for hydrogen embrittlement relief is within 4 hours after plating and is baked at 375° ± 25° F for a minimum of 4 hours	Visual
Major:		
	101. Overall length of bolt	Gage
	102. Bearing surface (min)	Gage
	103. Diameter of bolt head	Gage
	104. Diameter of bolt	Gage
	105. Diameter of roll pin hole	Gage
	106. Pitch diameter	Gage
	107. Major diameter	Gage
	108. Bolt head length (max)	Gage
	109. Max radius (0.125 - 0.060R)	Gage
	**110. Objective evidence of Radiographic Inspection	Visual*
	111. Out of roundness or flashing present	Gage
	**112. Objective evidence of magnetic particle inspection (before plating)	Visual*

*Verified Inspection Records/Objective Evidence supplied by contractor.
**100% Inspection

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112. Self locking element missing or not fused properly (no recesses) to thread.....Visual/Manual
113. Plating thickness (see 4.3.3.3).....Gage

Minor:
201. Evidence of poor workmanship (see 3.7).....Visual*

*Verified Inspection Records/Objective Evidence supplied by contractor.

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with dwg 8808932 or 8808362	Visual
	102. Marking not in accordance with dwg 8808932 or 8808362	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.7)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below: Tests b and c shall be performed on the same sample.

Test	Test Classification
a. Continuous grain flow lines	Major
b. Hardness	Major
c. Tensile load	Major
d. Plating thickness	Major

Of these tests, a and c are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Hardness. Each bolt in the sample (see Table III) shall be subjected to this test (see 4.4.1). If any bolt fails to comply with the applicable requirements, 3.3 it shall be cause for rejection of the entire lot.

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4.3.3.2 Tensile load. Each bolt in the sample (see Table III) shall be subjected to this test (see 4.3.3). If any bolt fails to comply with the applicable requirement 3.5, it shall be cause for rejection of the entire lot.

4.3.3.3 Plating thickness. Each bolt in the sample (if plating is done on one or two batches, a sample shall be taken from each batch being plated according to batch size and Table I. If plating is done in three or more batches, a sample shall be taken from each batch being plated according to batch size and Table III) shall be subjected to this test (see 4.4.2). If any bolt fails to comply with the applicable requirements in 3.4 it shall be cause for rejection of the entire batch.

4.3.3.4 Continuous grain flow lines. Three (3) bolts from each lot shall be randomly selected and subjected to this test (see 4.4.4). If any of these three bolts fails to comply with the requirement 3.7 it shall be cause for rejection of the entire lot.

Table III

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equpment, operating instructions and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attn: DRDAR-QAN-I as specified in DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests.

The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Hardness. Each bolt in the sample shall be placed in a suitable fixture and tested for a Rockwell hardness using standard test methods.

4.4.2 Plating thickness. Each bolt in the sample shall be tested for the thickness of the plating. The thickness of the plating will be determined by the use of an eddy current thickness tester.

4.4.3 Tensile load. A tensile load shall be applied to the test bolts using the test fixture shown in Figure 1. Apply a tensile load to the bolt using the test fixture on external nut wrenching 9237309 with a washer (MS15795-820) by applying a torque of 225 ft.-lbs. to the nut for 24 hours. This test is to be done on bolts which are not plated.

4.4.4 Continuous grain flow lines. Three (3) bolts in the sample shall be sectioned, (approximately in half) and etched using the following procedure. After the bolts are sectioned they shall be cleaned and polished using 200 emery paper. They shall then be placed in a warm acid bath of hydrochloric acid and water (approximately half and half) for about 5 to 10 minutes. After removal from the bath, they shall be cleaned and examined for the grain flow lines as shown in Figure 2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The bolt shall be packaged in accordance with dwg 8808932 using the specified moisture barrier.

5.1.2 Level C. The bolt shall be packaged in accordance with dwg 8808932 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The bolt shall be packed in accordance with dwgs 8808932 or 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8808932 or 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

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b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

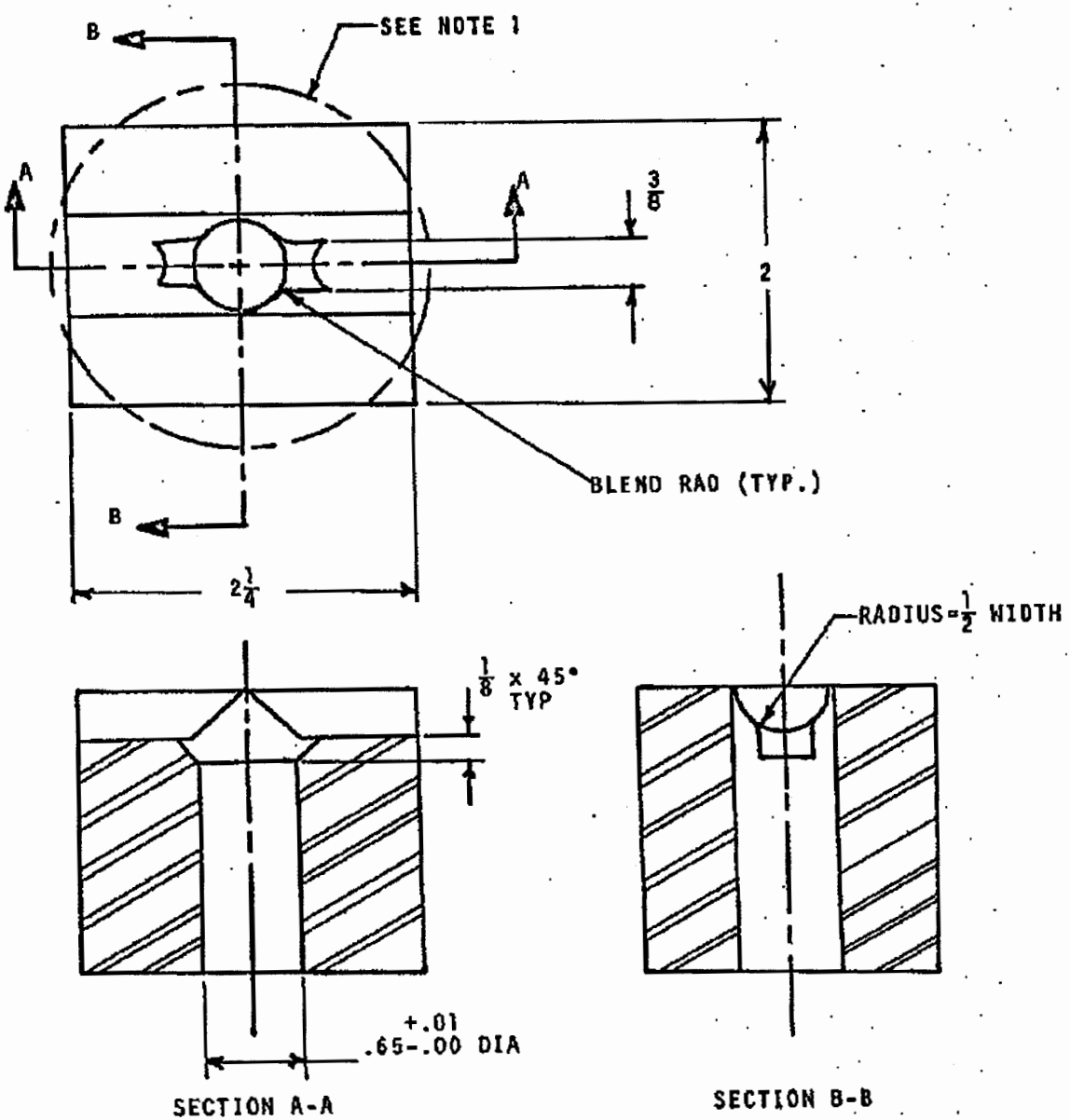
6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: DRDAR-QAN-I .

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No: NUOR-A-075

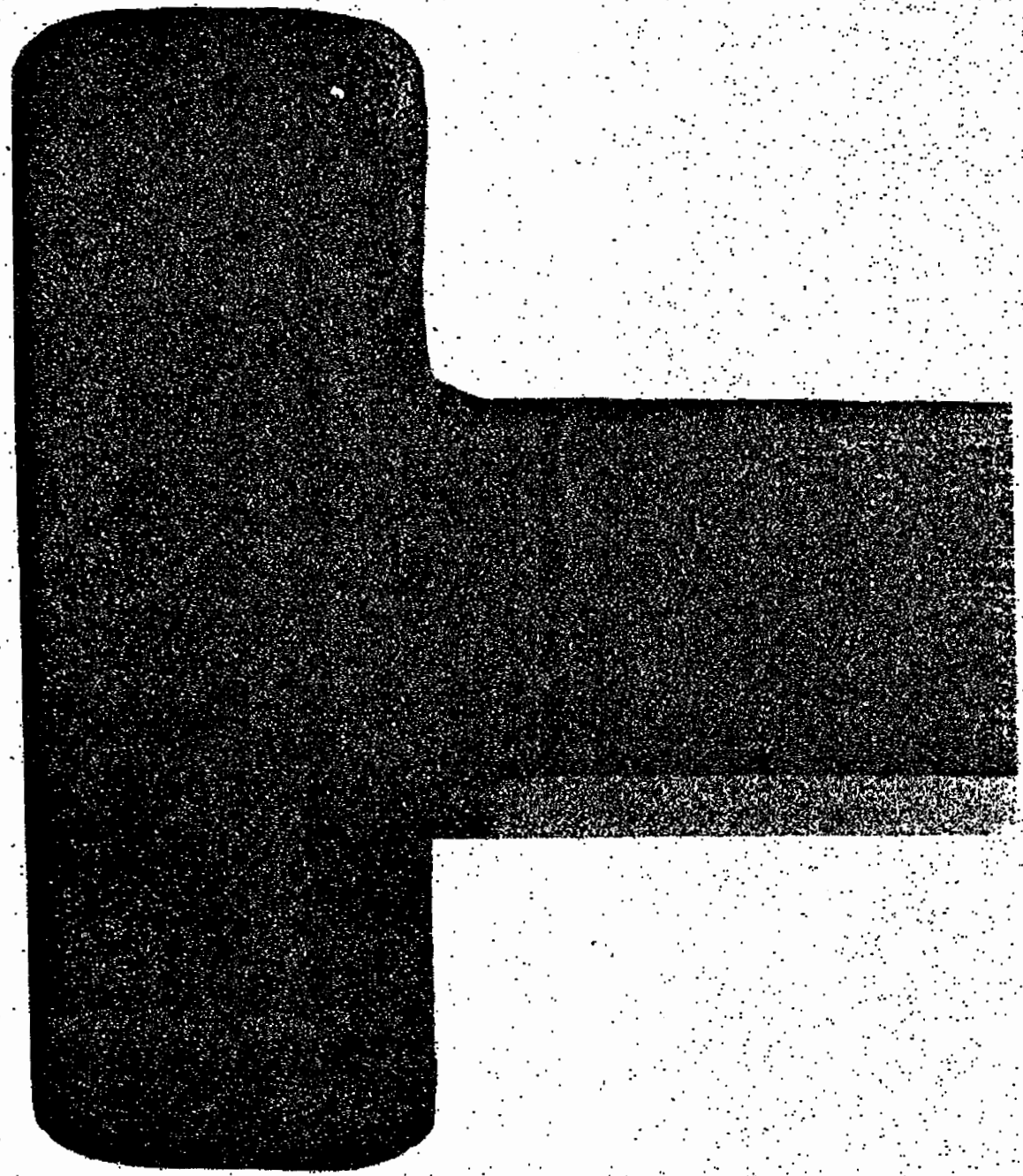
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NOTES: 1-ALTERNATE MAY BE 2-1/2 DIA BAR STOCK.
2-BREAK ALL SHARP EDGES 1/32 R APPROX.

Figure 1.

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TEE BOLT
GRAIN FLOW LINES
FIGURE 2

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